CLAUDE CODE FOR COURSE DEVELOPMENT

HOW TO BUILD A FULL GA COURSE IN GITHUB USING CLAUDE + CLAUDE CODE

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GENERAL ASSEMBLY · INTERNAL

GA TECHNICAL COURSE · 16 MODULES

01 CLAUDE VS CLAUDE CODE

CLAUDE (CLAUDE.AI)

- → Thinks and designs with you
- → Reads your course documents
- → Builds the CLAUDE.md blueprint
- → Iterates in conversation

CLAUDE CODE (TERMINAL)

- → Runs on your machine
- → Creates real files + folders
- → Pushes to GitHub directly
- → Follows CLAUDE.md exactly

> **The connection:** Claude designs the CLAUDE.md → Claude Code builds the course from it. You need both.

02 INSTALL CLAUDE CODE

Requires **Node.js 18+** — install via nodejs.org or Homebrew

Install: `npm install -g @anthropic-ai/claude-code`

Set API key: `export ANTHROPIC_API_KEY=sk-ant-...`

Launch in any folder: `claude`

GitHub push: add a Personal Access Token with repo scope

03 BUILD THE CLAUDE.MD — THE MOST IMPORTANT STEP

DO THIS IN CLAUDE (CLAUDE.AI) — NOT IN CLAUDE CODE

MODULE 1 Your example README + lab + **COMPETENCY MAP** .xlsx — LOs + lab descriptions + **SYLLABUS** .docx — course structure + **PACING GUIDE** .xlsx — timing per module → **CLAUDE.MD** The blueprint Your IP

WHAT CLAUDE.MD CONTAINS

- Language rules — American English
- Color palette, badge system, Mermaid styles
- Exact repo structure for every module

TEMPLATES INSIDE

- Module README template
- Lab README template
- SQL schema + seed data template

MODULE-LEVEL SPECIFICS

- AI Integration section per module
- AI Reflection prompt per module
- Tools per module (SQL → Spark)

04 REPOSITORY ARCHITECTURE

ONE GITHUB REPO PER MODULE

```
[N]-[module-slug]/
├── README.md ← badges, Mermaid, PNG, AI section
├── instructor-guide.md
├── assets/ ← matplotlib dark theme PNG
└── Lab-[lab-name]/
    ├── README.md ← lab brief + tasks
    ├── data/
    │   ├── schema_and_seed.sql
    │   └── lab_starter.sql
    ├── outputs/
    │   └── my_answers.md
```

05 BUILD LOCALLY → DEPLOY TO GITHUB

Create a folder for the course on your Mac — place CLAUDE.md inside

Open Claude Code in that folder: `cd my-course && claude`

Claude Code reads CLAUDE.md automatically — no extra setup

Prompt: **"Build all modules following CLAUDE.md"** — files appear locally

GITHUB DEPLOYMENT (WITH TOKEN)

Add GitHub token to Claude Code — repos push automatically

Prompt: **"Create repo for Module X and push"** — done in seconds

```
└── solutions/
    └── lab_solution.sql
```

06 WHAT THIS PRODUCED — A 16-MODULE GA TECHNICAL COURSE WITH LABS

16
Module READMEs with Mermaid + PNG + AI sections

16
Complete labs with SQL, seed data, solutions

Full Labs Audit
Less SME time consumed

Runs Labs. Detects & Fixes Bugs.
Automated smoke tests validate every lab before students see them

Claude + Claude Code · Not for distribution outside this team